

KL DIVERGENCE

KL(p||q) = \int p(x) \log \frac{p(x)}{q(x)} dx = \mathbb{E}_{x \sim p} \left[\log \frac{p(x)}{q(x)} \right]

KL ≥ 0; KL = 0 ⇔ p = q; asymmetric.
Univariate Gaussian KL (closed form):

KL[N(μ1, σ1)||N(μ2, σ2)] = log \frac{σ2}{σ1} + \frac{σ1^2 + (μ1 - μ2)^2}{2σ2^2} - \frac{1}{2}

VAE KL (diag, q vs. N(0, I)):

KL[N(μ, diag σ^2)||N(0, I)] = \frac{1}{2} \sum_j (\sigma_j^2 + \mu_j^2 - 1 - \log \sigma_j^2)

Multivariate Gaussian KL:

KL(N0||N1) = \frac{1}{2} \left[\log \frac{|\Sigma_1|}{|\Sigma_0|} + (\mu_0 - \mu_1)^T \Sigma_1^{-1} (\mu_0 - \mu_1) + \text{tr}(\Sigma_1^{-1} \Sigma_0) - D \right]

MLE = KL minimisation:

\hat{\theta} = \arg \max_{\theta} \sum_i \log p_{\theta}(\mathbf{x}_i) \Leftrightarrow \arg \min_{\theta} \text{KL}(p_{\text{data}} || p_{\theta})

DLVM / VAE

Generative model: z ~ p(z), x ~ p_{\theta}(x|z); intractable marginal p(x) = \int p_{\theta}(x|z)p(z)dz
ELBO (introduce q_{\phi}(z|x) > 0, apply Jensen log E ≥ E log):

\log p(\mathbf{x}) = \log \mathbb{E}_{\mathbf{z} \sim q_{\phi}} \left[\frac{p_{\theta}(\mathbf{x}|\mathbf{z})p(\mathbf{z})}{q_{\phi}(\mathbf{z}|\mathbf{x})} \right] \geq \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\phi}} [\log p_{\theta}(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\text{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p(\mathbf{z}))}_{\text{regularisation}} =: \mathcal{L}

Gap identity:

\log p(\mathbf{x}) = \mathcal{L}(\theta, \phi) + \text{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) || p_{\theta}(\mathbf{z}|\mathbf{x}))

Maximising ELBO maximises evidence and minimises KL to true posterior.

Amortised VI (encoder):

q_{\phi}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_{\phi}(\mathbf{x}), \text{diag } \boldsymbol{\sigma}_{\phi}^2(\mathbf{x}))

Reparameterisation trick:

\mathbf{z} = \boldsymbol{\mu}_{\phi}(\mathbf{x}) + \boldsymbol{\sigma}_{\phi}(\mathbf{x}) \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})

\nabla_{\phi} \mathcal{L} \approx \nabla_{\phi} \log p_{\theta}(\mathbf{x}|\boldsymbol{\mu}_{\phi} + \boldsymbol{\sigma}_{\phi} \odot \boldsymbol{\epsilon}) - \nabla_{\phi} \text{KL}

Low variance. Works for Gaussian/Beta/Gamma; not discrete (use Gumbel-softmax).

VAE Issues

Posterior collapse: q_{\phi}(\mathbf{z}|\mathbf{x}) \rightarrow p(\mathbf{z}), decoder ignores z. Fix: KL warm-up (\beta\text{-VAE: weight KL by } \beta \in (0, 1)).

Hole problem: aggregated posterior q_{\phi}(\mathbf{z}) = \frac{1}{N} \sum_n q_{\phi}(\mathbf{z}|\mathbf{x}_n) \neq p(\mathbf{z}) \Rightarrow \text{holes give bad samples.}

Improving the Prior

Aggregated posterior (best prior, minimises CE): q_{\phi}(\mathbf{z}) = \frac{1}{N} \sum_n q_{\phi}(\mathbf{z}|\mathbf{x}_n)

MoG prior: p_{\lambda}(\mathbf{z}) = \sum_{k=1}^K w_k \mathcal{N}(\mathbf{z}|\boldsymbol{\mu}_k, \text{diag } \boldsymbol{\sigma}_k^2)

VampPrior: p_{\lambda}(\mathbf{z}) = \frac{1}{K} \sum_{k=1}^K q_{\phi}(\mathbf{z}|\mathbf{u}_k), \quad \text{pseudo-inputs } \mathbf{u}_k \text{ learnable.}

Hierarchical VAE (2-level)

Bottom-up: Q(z1, z2|x) = q(z2|z1) q(z1|x)

\mathcal{L} = \mathbb{E}[\log p(\mathbf{x}|\mathbf{z}_1)] - \mathbb{E} \left[\log \frac{q(\mathbf{z}_1|\mathbf{x})}{p(\mathbf{z}_1|\mathbf{z}_2)} \right] - \mathbb{E}[\text{KL}(q(\mathbf{z}_2|\mathbf{z}_1) || p(\mathbf{z}_2))]

NORMALIZING FLOWS

Model: u ~ p_{\phi}(u), x = T(u); T diffeomorphism (bijective, smooth).

Change of variables:

p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{u}}(T^{-1}(\mathbf{x})) |\det J_{T^{-1}}(\mathbf{x})|

Inverse function theorem: J_{T^{-1}} = J_T^{-1}.

Log-likelihood (MLE):

\ell = \sum_i [\log p_{\phi}(T^{-1}(\mathbf{x}_i)) + \log |\det J_{T^{-1}}(\mathbf{x}_i)|]

Only need T^{-1}, J_{T^{-1}} for MLE; need T for sampling.

Composition T = T_K \circ \dots \circ T_1:

\log |\det J_T| = \sum_{k=1}^K \log |\det J_{T_k}(\mathbf{z}_{k-1})|

Affine Coupling Layer (RealNVP)

Partition z = (z1:d, zd+1:D); s, t : R^d -> R^{D-d} arbitrary NNs.

Fwd: z'1:d = z1:d, z'd+1:D = e^{s(z1:d)} \odot zd+1:D + t(z1:d)

Inv: z1:d = z'1:d, zd+1:D = (z'd+1:D - t) \odot e^{-s(z1:d)}

Jacobian (triangular => easy det):

J_T = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ * & \text{diag}(e^{s(\mathbf{z}_{1:d})}) \end{bmatrix}, \det J_T = \exp(\sum_i s_i)

Permutation layer: |det J| = 1; needed so all dims transform.

Flow as VAE Prior

\mathcal{L} = \mathbb{E}_{\mathbf{z} \sim q_{\phi}} [\log p_{\theta}(\mathbf{x}|\mathbf{z}) + \log p_{\phi}(T^{-1}(\mathbf{z}))] - \mathbb{E}[\log |\det J_T(T^{-1}(\mathbf{z}))|] - \mathbb{E}[\log q_{\phi}]

Prefer flow prior over flow posterior (more expressive model, same cost).

DDPM: FORWARD PROCESS

One step: $q(\mathbf{z}_t|\mathbf{z}_{t-1}) = \mathcal{N}(\mathbf{z}_t|\sqrt{1-\beta_t}\mathbf{z}_{t-1}, \beta_t\mathbf{I})$
Marginal (closed form, skips intermediate steps):
 $q(\mathbf{z}_t|\mathbf{x}) = \mathcal{N}(\mathbf{z}_t | \sqrt{\alpha_t}\mathbf{x}, (1-\alpha_t)\mathbf{I})$

where $\bar{\alpha}_t = \prod_{s=1}^t (1-\beta_s)$; $\alpha_t = 1-\beta_t$.
Reparametrisation: $\mathbf{z}_t = \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
Proof $q(\mathbf{z}_T|\mathbf{x}) \rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I})$: $\bar{\alpha}_T < (1-\beta_1)^T \xrightarrow{T \rightarrow \infty} 0 \Rightarrow \sqrt{\alpha_T} \rightarrow 0, 1-\bar{\alpha}_T \rightarrow 1$.

DDPM: REVERSE & ELBO

Reverse process (learned): $p_\theta(\mathbf{z}_{t-1}|\mathbf{z}_t) = \mathcal{N}(\mathbf{z}_{t-1}|\boldsymbol{\mu}_\theta(\mathbf{z}_t, t), \sigma_t^2\mathbf{I})$
Forward posterior (tractable! Gaussian via Bayes):

$$q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = \mathcal{N}(\mathbf{z}_{t-1}|\tilde{\boldsymbol{\mu}}_t, \tilde{\beta}_t\mathbf{I})$$
$$\tilde{\boldsymbol{\mu}}_t = \frac{\sqrt{\alpha_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x} + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{z}_t$$
$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

ELBO (hierarchical VAE — L_T ignorable, L_0 reconstruction):

$$\log p(\mathbf{x}) \geq \underbrace{-\text{KL}(q(\mathbf{z}_T|\mathbf{x})\|p(\mathbf{z}_T))}_{L_T \approx 0}$$
$$- \sum_{t=2}^T \underbrace{\mathbb{E}_{q(\mathbf{z}_t|\mathbf{x})}[\text{KL}(q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x})\|p_\theta)]}_{L_{t-1}}$$
$$+ \underbrace{\mathbb{E}[\log p(\mathbf{x}|\mathbf{z}_1)]}_{L_0}$$

L_{t-1} **closed form** (Gaussian KL, C indep. of θ):

$$L_{t-1} = \frac{1}{2\sigma_t^2} \|\tilde{\boldsymbol{\mu}}_t(\mathbf{z}_t, \mathbf{x}) - \boldsymbol{\mu}_\theta(\mathbf{z}_t, t)\|^2 + C$$

Noise prediction — substituting reparametrisation into $\tilde{\boldsymbol{\mu}}_t$:

$$\tilde{\boldsymbol{\mu}}_t = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \boldsymbol{\epsilon} \right) \Rightarrow \text{train } \boldsymbol{\epsilon}_\theta \approx \boldsymbol{\epsilon}$$

Simplified loss:

$$L_{t-1} \approx \|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}, t)\|^2$$

Training & Sampling

Train: $t \sim \mathcal{U}$; $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; $\mathbf{z}_t = \sqrt{\alpha_t}\mathbf{x} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}$;
gradient step on $\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t)\|^2$.
Sample: $\mathbf{z}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$; for $t = T, \dots, 1$; $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\mathbf{z}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{z}_t - \frac{\beta_t}{\sqrt{1-\alpha_t}} \boldsymbol{\epsilon}_\theta(\mathbf{z}_t, t) \right) + \sqrt{\beta_t} \boldsymbol{\eta}$$

SDE-BASED DIFFUSION (VP-SDE)

Forward SDE: $d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}dt + \sqrt{\beta(t)}d\mathbf{w}$
Reverse SDE (score $\mathbf{s} = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$):

$$d\mathbf{x} = [f(\mathbf{x}, t) - g(t)^2\mathbf{s}]dt + g(t)d\tilde{\mathbf{w}}$$

Probability flow ODE (same marginals, deterministic):

$$\frac{d\mathbf{x}}{dt} = f(\mathbf{x}, t) - \frac{1}{2}g(t)^2\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$$

Denosing Score Matching (train $\mathbf{s}_\theta \approx \nabla \log p$):

$$L_{\text{DSM}} = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_t} [\lambda(t) \|\mathbf{s}_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{x}_0)\|^2]$$

Linear SDE $\Rightarrow p(\mathbf{x}_t|\mathbf{x}_0)$ Gaussian $\Rightarrow$ tractable.
Key: $\nabla_{\mathbf{x}_t} \log \mathcal{N}(\mathbf{x}_t|\boldsymbol{\mu}, \sigma^2\mathbf{I}) = (\boldsymbol{\mu} - \mathbf{x}_t)/\sigma^2 = -\boldsymbol{\epsilon}/\sigma$

FID

Embed via Inception-v3; fit $\mathcal{N}(\boldsymbol{\mu}_r, \Sigma_r)$, $\mathcal{N}(\boldsymbol{\mu}_g, \Sigma_g)$:

$$\text{FID} = \|\boldsymbol{\mu}_r - \boldsymbol{\mu}_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r\Sigma_g)^{1/2})$$

Lower is better. Fréchet (Wasserstein-2) distance between Gaussians.

Model Comparison				
Model	Likelihood	Inv.	Bottleneck	Speed
VAE	Approx	No	Yes	Fast
Flow	Exact	Yes	No	Fast
DDPM	Approx	No	No	Slow
SBGM	~Ex.	No/Yes	No	Slow

MODULE 2: RIEMANNIAN GEOMETRY

Identifiability: p_θ identifiable $\Leftrightarrow \theta \mapsto p_\theta$ injective.
VAEs/NNs fail: reparametrise $\mathbf{z}$ without changing $p_\theta(\mathbf{x})$.
Manifold: $\mathcal{M} = f(\mathcal{Z})$, $f : \mathcal{Z} \subseteq \mathbb{R}^M \rightarrow \mathbb{R}^D$ smooth, $D > M$.

Immersed: J_f full rank $\forall \mathbf{z}$. (Embedded $\Rightarrow$ immersed, not vice versa.)

MLP with full-rank weights + smooth activations $\Rightarrow$ immersed.

Pullback Riemannian metric:

$$M(\mathbf{z}) = J_f(\mathbf{z})^\top J_f(\mathbf{z}) \in \mathbb{R}^{M \times M}$$

Inner product: $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{z}} = \mathbf{u}^\top M(\mathbf{z})\mathbf{v}$.
Invariant to reparametrisations $\Rightarrow$ identifiable distances.

Curve length:

$$L(c) = \int_0^1 \sqrt{\dot{c}^\top M(c) \dot{c}} dt$$

Curve energy (Cauchy-Schwarz $\Rightarrow L(c)^2 \leq E(c)$, equality iff const. speed):

$$E(c) = \int_0^1 \dot{c}^\top M(c) \dot{c} dt$$

Constant-speed geodesic: energy minimiser with $\|\dot{c}\|_M = \text{const.}$

$$L(c)^2 \leq E(c) \text{ (C-S), equality iff const. speed}$$

$$\text{Geodesic ODE: } \ddot{c}_i + \sum_{j,k} \Gamma_{jk}^i \dot{c}_j \dot{c}_k = 0$$

$\Gamma_{jk}^i = \frac{1}{2} M^{im} (\partial_j M_{mk} + \partial_k M_{mj} - \partial_m M_{jk})$ (Christoffel symbols).

Boundary value (start+end) $\rightarrow$ may not be unique.

Initial value (point+velocity) $\rightarrow$ unique (Picard-Lindelöf).

Log map (Riemannian subtraction):

$\text{Log}_p(q)$ = initial velocity of geodesic $p \rightarrow q$ (tangent vector at p); $\|\text{Log}_p(q)\|_{M(p)} = \text{dist}(p, q)$.

Exp map (Riemannian addition):

$\text{Exp}_p(\mathbf{v})$ = endpoint of geodesic from p with init. velocity $\mathbf{v}$.

Noisy Manifold (Stochastic Decoder)

$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_\theta(\mathbf{z}), \text{diag } \boldsymbol{\sigma}_\theta^2(\mathbf{z}))$; rewrite as random projection $\mathbf{x} = \boldsymbol{\mu}_\theta(\mathbf{z}) + \Lambda_\theta^{1/2} \boldsymbol{\eta}$.

$$M(\mathbf{z}) = J_\mu^\top J_\mu + \underbrace{\mathbb{E}[J_{\Lambda^{1/2}}^\top J_{\Lambda^{1/2}}]}_{\text{uncertainty term}}$$

Uncertainty creates “walls” $\Rightarrow$ geodesics follow data.
Ensemble: K decoders; uncertainty = variance across models.

Information Geometry (Fisher-Rao)

KL is locally quadratic in $d\theta$ (2nd-order Taylor):

$$\text{KL}(p(\cdot; \theta) \| p(\cdot; \theta + d\theta)) \approx \frac{1}{2} d\theta^\top F(\theta) d\theta$$

$$F(\theta)_{ij} = \mathbb{E}[\partial_i \log p \cdot \partial_j \log p] \text{ (Fisher info. matrix)}$$

Fisher-Rao metric = Riemannian metric on distribution space; reparametrisation invariant.

Parameter Manifolds

$G(\theta) = \mathbb{E}_{\mathbf{x}}[J_f(\mathbf{x}; \theta)^\top J_f(\mathbf{x}; \theta)]$

Kernel manifold: zero-length curves (same function, diff. params).

Image manifold: positive-length curves (diff. functions).
UQ: project Gaussian param samples onto kernel $\Rightarrow$ interpolates data.

GRAPH DEFINITIONS

$G = (V, E)$; adjacency $A \in \{0, 1\}^{N \times N}$; degree $D = \text{diag}(d_1, \dots, d_N)$.

Laplacians:

$$L = D - A, \quad L_{\text{sym}} = D^{-\frac{1}{2}} L D^{-\frac{1}{2}} = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}}$$
$$L_{\text{rw}} = D^{-1} L = I - D^{-1} A$$

All non-negative eigenvalues.

GRAPH STATISTICS

Eigenvector Centrality

$$\lambda \mathbf{e} = A \mathbf{e} \quad e_u = \frac{1}{\lambda} \sum_{v \in \mathcal{N}(u)} e_v$$

Perron-Frobenius: largest eigenvalue is unique; eigenvector non-negative.

Clustering Coefficient

$$c_u = \frac{|\{(v_1, v_2) \in E : v_1, v_2 \in \mathcal{N}(u)\}|}{\binom{d_u}{2}}$$

$\mathbf{c} = (D(D - I))^{-1} \text{diag}(A^3)$; $c_u = [A^3]_{uu} / (d_u(d_u - 1))$.
Weisfeiler-Lehman (WL) Test

- 1. Init: $\ell_v^{(0)} = d_v$ (degree)
- 2. Update: $\ell_v^{(i)} = \text{hash}(\{\{\ell_u^{(i-1)} : u \in \mathcal{N}(v)\}\})$
- 3. Summarise: $\text{hash}(\{\{\ell_v^{(i)} : v \in V\}\})$
- 4. Different summaries $\Rightarrow$ not isomorphic (can fail for some pairs).

MESSAGE PASSING

$$\mathbf{m}_u = \text{agg}^{(k)}(\{\mathbf{h}_v^{(k)} : v \in \mathcal{N}(u)\})$$

$$\mathbf{h}_u^{(k+1)} = \text{update}^{(k)}(\mathbf{h}_u^{(k)}, \mathbf{m}_u)$$

Basic GNN:

$$\mathbf{h}_u^{(k)} = \sigma(W_s \mathbf{h}_u^{(k-1)} + W_n \sum_{v \in \mathcal{N}(u)} \mathbf{h}_v^{(k-1)} + \mathbf{b})$$

Degree-normalised: $\mathbf{m} = \sum_v \mathbf{h}_v / |\mathcal{N}(u)|$ or $\sum_v \mathbf{h}_v / \sqrt{|\mathcal{N}(u)| |\mathcal{N}(v)|}$

Set pooling (universal): $\mathbf{m} = \text{MLP}_\theta(\sum_v \text{MLP}_\phi(\mathbf{h}_v))$

Attention: $\alpha_{u,v} = \text{softmax}(\mathbf{h}_u^\top W \mathbf{h}_v)$, $\mathbf{m} = \sum_v \alpha_{u,v} \mathbf{h}_v$

Scaled dot-product: $\alpha_{u,v} = \text{softmax}(\mathbf{k}_u^\top \mathbf{q}_v / \sqrt{d})$, $\mathbf{m} = \sum_v \alpha_{u,v} \mathbf{v}_v$

GRU update (reduce over-smoothing):

$$\mathbf{r} = \sigma(W_{mr} \mathbf{m} + W_{hr} \mathbf{h} + \mathbf{b}_r)$$

$$\mathbf{z} = \sigma(W_{mz} \mathbf{m} + W_{hz} \mathbf{h} + \mathbf{b}_z)$$

$$\tilde{\mathbf{h}} = \tanh(W_{mh} \mathbf{m} + W_{hh} (\mathbf{r} \odot \mathbf{h}) + \mathbf{b}_h)$$

$$\mathbf{h}^{\text{new}} = \mathbf{z} \odot \tilde{\mathbf{h}} + (1 - \mathbf{z}) \odot \mathbf{h}$$

Residual: $\mathbf{h}^{\text{new}} = \mathbf{h} + \text{update}(\mathbf{h}, \mathbf{m})$

NODE EMBEDDINGS (SHALLOW)

Encoder (lookup table): $\text{enc}(u) = \mathbf{z}_u = Z^\top \mathbf{s}_u$ ($\mathbf{s}_u$ one-hot). Transductive only.

Decoders: $\mathbf{z}_u^\top \mathbf{z}_v$; $\sigma(\mathbf{z}_u^\top \mathbf{z}_v + b)$; $e^{\mathbf{z}_u^\top \mathbf{z}_v} / \sum_w e^{\mathbf{z}_u^\top \mathbf{z}_w}$;
 $\|\mathbf{z}_u - \mathbf{z}_v\|^2$

MSE loss:

$$L_{\text{MSE}} = \sum_{(u,v)} (S_{uv} - \mathbf{z}_u^\top \mathbf{z}_v)^2 = \|S - ZZ^\top\|_F^2$$

Binary cross-entropy (note $1 - \sigma(x) = \sigma(-x)$):

$$L_{\text{BCE}} = - \sum_{(u,v) \in E} \log \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b) - \sum_{(u,v) \notin E} \log \sigma(-\mathbf{z}_u^\top \mathbf{z}_v - b)$$

Random walk (neg. sampling):

$$L_{\text{RW}} = \sum_{(u,v) \in W} \left[-\log \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b) - \gamma \mathbb{E}_w [\log \sigma(-\mathbf{z}_u^\top \mathbf{z}_w - b)] \right]$$

GRAPH FOURIER & CONVOLUTIONS

Graph convolution (adjacency as shift):

$$\mathbf{y} = \left(\sum_{k=0}^K h[k] A^k \right) \mathbf{x}$$

DFT: $\tilde{x}[k] = \sum_n x[n] \omega_N^{kn}$, $\omega_N = e^{-i2\pi/N}$
Matrix form: $\tilde{\mathbf{x}} = U^H \mathbf{x}$, $\mathbf{x} = U \tilde{\mathbf{x}}$, U unitary ($U^H U = I$).
Convolution theorem (circ. conv. $\Leftrightarrow$ freq. mult.):

$$\mathbf{y} = H \mathbf{x} \Leftrightarrow \tilde{\mathbf{y}} = \tilde{\mathbf{h}} \odot \tilde{\mathbf{x}}$$

Proof:

$$\begin{aligned} \tilde{y}[k] &= \sum_n \sum_\ell x[\ell] h[(n-\ell) \bmod N] \omega_N^{kn} \\ &= \sum_\ell x[\ell] \omega_N^{k\ell} \underbrace{\sum_m h[m] \omega_N^{km}}_{\tilde{h}[k]} = \tilde{x}[k] \tilde{h}[k] \end{aligned}$$

Graph Fourier: $A = U \Lambda U^H$. Cycle graph eigenvectors = DFT basis.

Graph conv. via Fourier:

$$\mathbf{y} = U \left(\sum_{k=0}^K h[k] \Lambda^k \right) U^H \mathbf{x}$$

GNNs AND PROBABILISTIC MODELS

Markov Random Field:

$$p(\{x_v\}, \{z_v\}) \propto \prod_v \Phi(x_v, z_v) \prod_{(u,v) \in E} \Psi(z_u, z_v)$$

Mean field VI (factored $q = \prod_v q_v$, min KL — fixed point):

$$\log q_v^{(t+1)} = c_v + \log \Phi(x_v, z_v) + \sum_{u \in \mathcal{N}(v)} \int q_u^{(t)} \log \Psi dz_u$$

This IS message passing; kernel embeddings $\Rightarrow$ GNN updates.

GRAPH VAE

Node-level VAE (encoder = GNN):

$$\mu_Z, \log \sigma_Z = \text{GNN}(A, X), \quad Z = \mu_Z + \varepsilon \odot \sigma_Z$$

$$P(A_{uv} = 1|Z) = \sigma(\mathbf{z}_u^\top \mathbf{z}_v + b), \quad p(Z) = \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$\mathcal{L} = \mathbb{E}_q[\sum_{u,v} \log p_\theta(A_{uv}|Z)] - \text{KL}[q_\phi(Z|G) \| p(Z)]$$

Graph-level VAE: $\mu_z, \log \sigma_z = \text{GNN_readout}(A, X)$;
 $z = \mu_z + \varepsilon \sigma_z$; $\tilde{A} = \sigma(\text{MLP}(z))$.
GAN: $\min_\theta \max_\phi \mathbb{E}_x [\log(1 - d_\phi(x))] + \mathbb{E}_z [\log d_\phi(g_\theta(z))]$
Generator: $\tilde{A} = \sigma(\text{MLP}(z))$; Discriminator: permutation-invariant GNN.

SYMMETRY IN GEOMETRIC GNNs

Invariant: $f(T(\mathbf{h})) = f(\mathbf{h})$ **Equivariant:** $f(T(\mathbf{h})) = T'(f(\mathbf{h}))$
Inv. ops: $\|\mathbf{v}\|$, $\mathbf{v}_1 \cdot \mathbf{v}_2$, $\|\mathbf{v}_1 \times \mathbf{v}_2\|$ Equiv. ops: $a\mathbf{v}$, $\mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{v}_1 \times \mathbf{v}_2$

EVALUATION OF GRAPH GENERATION

Total variation: $d(s_{\text{gen}}, s_{\text{test}}) = \sup_x |P(s_{\text{gen}} \in x) - P(s_{\text{test}} \in x)|$
Discrete: $d = \frac{1}{2} \|p_{\text{gen}} - p_{\text{test}}\|_1$. Compare: degree, clust. coeff., centrality.

Traditional models:

- **Erdős-Rényi:** $P(A_{uv} = 1) = r$ (edges i.i.d.)
- **Stochastic block model:** nodes $\rightarrow$ blocks; $P(A_{uv} = 1) = r_{ij}$
- **Preferential attachment:** $P(A_{uv} = 1) \propto d_v^{(t)}$ (rich-get-richer)

KEY IDENTITIES

Jensen: concave g : $g(\mathbb{E}[X]) \geq \mathbb{E}[g(X)]$ (e.g. $g = \log$)
Cauchy-Schwarz: $(\int f \cdot g)^2 \leq (\int f^2)(\int g^2)$; equality iff $f \propto g \Rightarrow L(c)^2 \leq E(c)$
PPCA bias: $\hat{\mathbf{b}} = \bar{\mathbf{x}}$ (set $\partial \ell / \partial \mathbf{b} = 0 \Rightarrow C^{-1}(\mathbf{b} - \bar{\mathbf{x}}) = 0$)
Triangular det: $\det A = \prod_i a_{ii}$
Jacobian chain rule: $J_{T \circ S}(x) = J_T(S(x)) \cdot J_S(x)$
Gaussian score: $\nabla_{\mathbf{x}} \log \mathcal{N}(\mathbf{x}|\mu, \sigma^2 I) = (\mu - \mathbf{x})/\sigma^2 = -\varepsilon/\sigma$
LOTUS: $\mathbb{E}[f(X)] = \int f(x)p(x)dx$ (sum over $t \rightarrow$ expectation $\sim \mathcal{U}$)
Bayes on forward: $q(\mathbf{z}_{t-1}|\mathbf{z}_t, \mathbf{x}) = q(\mathbf{z}_t|\mathbf{z}_{t-1})q(\mathbf{z}_{t-1}|\mathbf{x})/q(\mathbf{z}_t|\mathbf{x})$ (Gaussian, closed form)
Transduction vs induction: Shallow = transductive. GNN = inductive.